curriculum vitae

Computational Engineer with a Ph.D. in Computational & Experimental Mechanics and deep expertise in scientific computing, numerical methods, and high-performance Python tooling. Specialized in designing, optimizing, and deploying scalable surrogate models and data pipelines that bridge low length-scale physics simulations with practical engineering applications. Proven track record developing production-quality scientific software, including Python-to-C++ integrations, interactive visualization tools, and HPC workflows, that accelerate research and improve team productivity. Strong background in algorithm development, performance optimization, verification & validation, and turning complex numerical problems into robust, reusable solutions.

Personalia

Surname:	Ruybalid
Given names	Andre, Paul
Degrees:	Ph.D. | MSc. | BSc.
City:	Los Alamos, New Mexico, U.S.A.
Phone number:	+1 719 210 1746 (personal mobile)
E-mail:	andreruybalid@gmail.com
Date of birth:	26 October 1987
Place of Birth:	Farmington, New Mexico, U.S.A.
Nationality:	American/Dutch

Professional experience

2023 - Present — **R&D Engineer, Los Alamos National Laboratory**

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

- ❖ Developed multiple surrogate models and custom Python-based tools to bridge polycrystalline mechanics to engineering-scale predictions, reaching R&D 100 Award Finals twice.
- ❖ Designed and executed large-scale simulation pipelines generating thousands of low-length-scale simulations to train data-driven surrogate models for viscoplastic creep behavior.
- ❖ Ported and integrated Python-based data-driven surrogate model inference into MOOSE/BISON (C++ framework), designing efficient C++ implementations while maintaining clean Python interfaces for numerical workflows. Contributed to model verification, validation, and uncertainty quantification pipelines.
- ❖ Created Python tooling for data analysis, visualization, and automation of complex multi-scale modeling workflows.
- ❖ Coordinated with mechanistic modelers to integrate surrogate models into larger simulation frameworks for industrial and national lab partners.

2021 – 2023 — **Postdoctoral Researcher, Los Alamos National Laboratory**

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

- ❖ Developed the surrogate model ("LAROMance") for polycrystalline nuclear reactor materials by designing large-scale simulation pipelines and curating datasets from thousands of low length-scale simulations.
- ❖ Implemented physics-informed machine learning techniques and numerical algorithms to capture microstructure-dependent material behavior.

2019 – 2020 — **Design Engineer, ASML**

DUV Def & Metrology group of Defectivity Department | Eindhoven, The Netherlands

Developed and maintained complex defect metrology algorithms in custom MATLAB tooling for high-precision nanoscale defect detection on DUV lithography reticles. Designed, validated, and optimized numerical algorithms through cleanroom experimentation and large-scale data analysis, delivering results under tight deadlines in collaboration with cross-functional and industrial teams.

2013 – 2019 **PhD Researcher, Eindhoven University of Technology**

Computational and Experimental Mechanics | Prof. Dr. Marc Geers & Dr. Johan Hoefnagels.
Philips Research, Materials Innovation Institute | Eindhoven, The Netherlands

- Designed and implemented an integrated metrology framework combining finite element modeling, custom micromechanical instrumentation, and in-situ microscale experimentation, resulting in five peer-reviewed journal publications.
- Developed inverse optimization algorithms, digital image correlation methods, and data processing pipelines for full-field identification of constitutive parameters using high-resolution microscopy.

2017 – 2020 **Account manager & Technical recruiter**

Research-Square & EuFlex | Eindhoven, The Netherlands

Acquired and managed client accounts, and conducted technical recruitment of highly specialized engineering and PhD-level candidates for science and technology companies.

2018 – 2019 **Lab technologist, Multi-Scale Laboratory**

Eindhoven University of Technology | Eindhoven, The Netherlands
Computational and Experimental Mechanics

Managed an experimental mechanics laboratory, overseeing high-tech equipment including scanning electron microscopes and micro-mechanical test systems, while providing technical guidance to researchers and students and supporting experimental design in a high-demand environment.

2009 – 2013 **Tutor of students**

Eindhoven University of Technology, Mechanical Engineering Department | Freelance

Mentored bachelor student teams on engineering case studies, supporting balanced group dynamics, resolving team issues, and delivering constructive feedback.

Education

Eindhoven University of Technology, the Netherlands
Department of Mechanical Engineering
Division: Computational and Experimental Mechanics
Research Group: Mechanics of Materials

2013 – 2019 **PhD.-research project**
Title: "Identification of multi-layer interface systems: a full-field, micro-mechanical approach."

2013 – 2019 **Graduate School of Engineering Mechanics**
Advanced Mechanics Courses

2010 – 2013 **MSc. Mechanical Engineering**
Minor: Biomedical Engineering

2006 – 2010 **BSc. Mechanical Engineering**

2000 – 2006 **Research-oriented secondary education (VWO)**
St.-Janscollege, Hoensbroek, The Netherlands

Selected journal publications (peer-reviewed)

P. Robbe, A.P. Ruybalid, A. Hegde, C. Bonneville, H.N. Najm, L. Capolungo and C. Safta . A comparison of surrogate constitutive models for viscoplastic creep simulation of HT-9 steel. *Computational Materials Science*, 262, 114367 (2026). https://doi.org/10.1016/j.commatsci.2025.114367.

A.P. Ruybalid, A. Tallman, W. Wen, C. Matthews, L. Capolungo. Data-Driven Surrogate Modeling with Microstructure-Sensitivity of Viscoplastic Creep in Grade 91 Steel. Integrating Materials and Manufacturing Innovation 13, 895–914 (2024). https://doi.org/10.1007/s40192-024-00377-z

A.P. Ruybalid, O. van der Sluis, M.G.D. Geers and J.P.M. Hoefnagels. Full-field identification of mixed-mode adhesion properties in a flexible, multi-layer microelectronic material system. *Engineering Fracture Mechanics*, 226, pp. 106879 (2020), *https://doi.org/10.1016/j.engfracmech.2020.106879*.

A.P. Ruybalid, O. van der Sluis, M.G.D. Geers, J.P.M. Hoefnagels. An *In-Situ*, Micro-Mechanical Setup with Accurate, Tri-Axial, Piezoelectric Force Sensing and Positioning. *Exp Mech* 60, 713–725 (2020), *https://doi.org/10.1007/s11340-020-00583-8*.

A.P. Ruybalid, J.P.M. Hoefnagels, O. van der Sluis, M.P.F.H.L. van Maris and M.G.D. Geers. Mixed-mode cohesive zone parameters from integrated digital image correlation on micrographs only. *International Journal of Solids and Structures*, 156-157, pp. 179-193 (2019), *https://doi.org/10.1016/j.ijsolstr.2018.08.010*.